Statistical Analysis Log

Diabetes 130-US Hospitals (1999-2008) | Readmission Analysis | Cleaned dataset: 101,763 rows x 47 columns

This log records every formal statistical test run against the cleaned dataset: the hypotheses tested, the method used, the exact statistic and p-value obtained, the effect size, and the resulting interpretation. It supplements (and does not replace) the exploratory data analysis and cleaning steps performed earlier in the notebook. Significance threshold used throughout: **alpha = 0.05**.

0. Target Variable Check

Readmission Category	Count	% of Total
No Readmission (0)	54,861	53.9%
Readmitted >30 Days (1)	35,545	34.9%
Readmitted <30 Days (2)	11,357	11.2%

The target is imbalanced (No Readmission is the majority class). This is noted because it affects how p-values should be read: with ~100k records, even small, practically unimportant differences will often test as statistically significant.

1. Chi-Square Test of Independence (Categorical Variables vs. Readmission)

H0: The variable is independent of readmission status. H1: The variable is associated with readmission status. Effect size reported as Cramer's V (0-1 scale; <0.1 negligible, 0.1-0.3 weak, 0.3-0.5 moderate, >0.5 strong).

Variable	Chi-square	df	p-value	Cramer's V	Sig.(a=.05)
insulin	516.43	6	2.42e-108	0.0504	Yes
diabetesMed	386.44	2	1.22e-84	0.0616	Yes
change	215.94	2	1.29e-47	0.0461	Yes
race	81.73	8	2.19e-14	0.02	Yes
gender	34.9	2	2.64e-08	0.0185	Yes
max_glu_serum	35.11	4	4.41e-07	0.0573	Yes
A1Cresult	16.78	4	0.0021	0.0222	Yes

Result: All seven categorical variables tested are statistically significant (p < 0.05), but every Cramer's V is below 0.1 except diabetesMed (0.062), change (0.046) and insulin (0.050) — all still in the negligible-to-weak range. **Interpretation:** gender, race, medication status, and lab results are technically associated with readmission at this sample size, but none of them is a practically strong predictor on its own.

2. Kruskal-Wallis H-Test (Numeric Variables vs. Readmission)

H0: The distribution (median) of the variable is the same across all three readmission groups. H1: At least one group differs. Used instead of one-way ANOVA because several numeric variables are heavily right-skewed (see skewness table in the EDA section), violating ANOVA's normality assumption. Effect size reported as epsilon-squared (same 0-1 scale as Cramer's V).

Variable	H statistic	p-value	Epsilon-sq.	Sig.(a=.05)
number_emergency	1651.7	< 1e-300	0.0162	Yes
number_inpatient	5661.69	< 1e-300	0.0556	Yes
number_outpatient	1251.88	1.44e-272	0.0123	Yes
number_diagnoses	1209.64	2.14e-263	0.0119	Yes
num_medications	547.64	1.21e-119	0.0054	Yes
time_in_hospital	469.07	1.39e-102	0.0046	Yes
num_procedures	234.87	9.96e-52	0.0023	Yes
num_lab_procedures	173.34	2.29e-38	0.0017	Yes
age	98.17	4.82e-22	0.0009	Yes

Result: All nine numeric variables are statistically significant. **number_inpatient** has by far the largest effect size (epsilon-sq. = 0.056, small-to-moderate), followed by number_emergency (0.016) and number_outpatient (0.012). The remaining variables (age, time_in_hospital, lab/medication counts) have epsilon-squared under 0.01 — negligible in practical terms despite significant p-values. **Interpretation:** prior inpatient visits is the numeric variable most worth attention as a readmission signal; the rest move the needle only slightly.

3. Post-Hoc Pairwise Comparisons (Mann-Whitney U, Bonferroni-corrected)

Run only for the 9 numeric variables above (all were significant), to identify which pair of readmission groups differs. Bonferroni correction adjusts for running 3 pairwise tests per variable.

Variable	Group A	Group B	p (Bonferroni)	Sig.
age	No	>30 Days	3.05e-14	Yes
age	No	<30 Days	5.50e-15	Yes
age	>30 Days	<30 Days	0.0124	Yes
time_in_hospital	No	>30 Days	2.88e-46	Yes
time_in_hospital	No	<30 Days	4.78e-83	Yes
time_in_hospital	>30 Days	<30 Days	3.49e-21	Yes
num_lab_procedures	No	>30 Days	1.18e-29	Yes
num_lab_procedures	No	<30 Days	3.49e-20	Yes
num_lab_procedures	>30 Days	<30 Days	0.2969	No
num_procedures	No	>30 Days	1.94e-51	Yes
num_procedures	No	<30 Days	1.21e-08	Yes
num_procedures	>30 Days	<30 Days	8.52e-05	Yes
num_medications	No	>30 Days	1.36e-75	Yes

Variable	Group A	Group B	p (Bonferroni)	Sig.
num_medications	No	<30 Days	1.07e-76	Yes
num_medications	>30 Days	<30 Days	1.30e-10	Yes
number_outpatient	No	>30 Days	2.18e-253	Yes
number_outpatient	No	<30 Days	1.13e-94	Yes
number_outpatient	>30 Days	<30 Days	0.0248	Yes
number_emergency	No	>30 Days	7.65e-288	Yes
number_emergency	No	<30 Days	4.99e-215	Yes
number_emergency	>30 Days	<30 Days	1.16e-05	Yes
number_inpatient	No	>30 Days	< 1e-300	Yes
number_inpatient	No	<30 Days	< 1e-300	Yes
number_inpatient	>30 Days	<30 Days	7.81e-65	Yes
number_diagnoses	No	>30 Days	4.69e-214	Yes
number_diagnoses	No	<30 Days	1.27e-111	Yes
number_diagnoses	>30 Days	<30 Days	0.1649	No

Result: Nearly every pair of readmission groups differs significantly for every variable. The two exceptions are num_lab_procedures (>30 Days vs <30 Days, p=0.297) and number_diagnoses (>30 Days vs <30 Days, p=0.165) — for these two variables, the >30-day and <30-day readmission groups are not distinguishable from each other, even though both differ from the No-Readmission group.

4. Spearman Correlation (Numeric Variables vs. Each Other)

H0: rho = 0 (no monotonic relationship). H1: rho != 0. Top 15 pairs by absolute correlation strength shown below; full pairwise set (36 pairs) was computed.

Variable A	Variable B	Spearman rho	p-value	Sig.(a=.05)
time_in_hospital	num_medications	0.465	< 1e-300	Yes
num_procedures	num_medications	0.352	< 1e-300	Yes
time_in_hospital	num_lab_procedures	0.337	< 1e-300	Yes
num_medications	number_diagnoses	0.294	< 1e-300	Yes
num_lab_procedures	num_medications	0.252	< 1e-300	Yes
time_in_hospital	number_diagnoses	0.237	< 1e-300	Yes
number_emergency	number_inpatient	0.222	< 1e-300	Yes
age	number_diagnoses	0.196	< 1e-300	Yes
time_in_hospital	num_procedures	0.187	< 1e-300	Yes
number_outpatient	number_emergency	0.177	< 1e-300	Yes
num_lab_procedures	number_diagnoses	0.169	< 1e-300	Yes
number_outpatient	number_inpatient	0.156	< 1e-300	Yes
number_inpatient	number_diagnoses	0.136	< 1e-300	Yes
age	time_in_hospital	0.12	< 1e-300	Yes
number_outpatient	number_diagnoses	0.113	3.67e-286	Yes

Result: The strongest relationship is time_in_hospital vs. num_medications (rho = 0.465, moderate), followed by num_procedures vs. num_medications (0.352) and time_in_hospital vs. num_lab_procedures (0.337). All remaining pairs are weak (rho < 0.30). **Interpretation:** longer hospital stays come with more medications and more lab work, as expected, but no pair is strongly redundant — multicollinearity is not a concern if these variables are used together later (e.g., in a predictive model).

5. Overall Summary

Every variable tested — all 7 categorical and all 9 numeric — came back statistically significant against readmission status. Given the sample size (~101,700 encounters), this is expected and is not, by itself, informative. What matters is effect size:

• **Most influential:** number_inpatient (epsilon-sq. = 0.056), and to a lesser extent number_emergency and number_outpatient — prior hospital utilization is the strongest available signal.
• **Weak but real:** diabetesMed, change, and insulin status (Cramer's V 0.05-0.06) show a measurable but small association with readmission.
• **Negligible:** gender, race, A1Cresult, age, time_in_hospital, and the remaining lab/medication counts (all effect sizes < 0.02) — statistically detectable, practically minor.
• **No multicollinearity concern:** numeric predictors are weakly correlated with each other (strongest rho = 0.465), so they can be used together in downstream modeling.

This confirms, with formal tests, the pattern the earlier EDA section suggested visually: no single demographic or utilization variable dominates readmission risk; it is driven by a combination of factors, most notably a patient's recent history of inpatient/emergency/outpatient visits.

Note: p-values of exactly 0 in scipy output (below float64 precision) are reported as '< 1e-300' rather than 0 to avoid implying an impossible exact-zero probability.